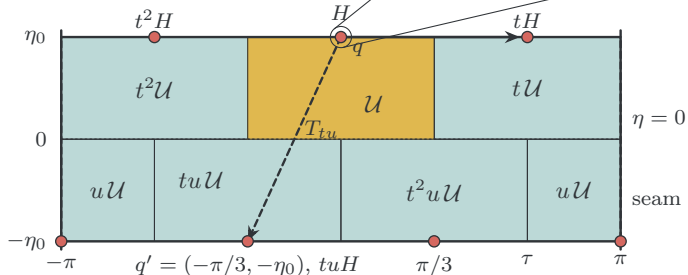
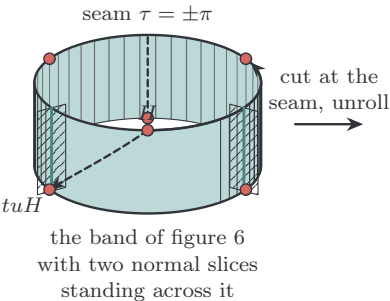
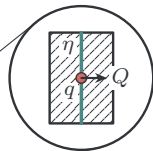


the slice at q :
the cut face of
figure 6



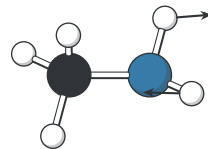
the chart, seam identified: six H -cells, $\mathcal{U}$ in gold, the same points and paths; $\eta = 0$ is the planar amino locus; T_{tu} carries q to q'

$$X_0(q)P_{tu} = A_{tu}X_0(q'), \quad A_{tu} = R_z(\pi)$$

$$T_{tu} : (r, \tau, \eta, Q_+, Q_-) \mapsto (rR_z(\pi), \tau - \frac{\pi}{3}, -\eta, Q_+, -Q_-) \quad X = r(X_0(q) + \sum_a Q_a e_a(q))$$

$$M_{tu} = \text{diag}(1, -1) \text{ on the normal directions } (e_+, e_-)$$

$e_{\pm}(q)$ orthogonal to the family and to the rotations in the mass metric



$e_-(q)$, the amino twist, arrows $3 \times$
the displacement $Q_- = 0.42 \text{ \AA}$